

Solving Inequalities in Real-World Contexts

Teacher's Edition

Standard and Targets

MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

Learning target I can write an inequality from a real-world situation, solve it, and say what the solution means in terms of the situation.

Essential questions How do you write an inequality from a context? How do you decide which way to round a boundary, and how do you check that you were right?

Building on Lessons 8 and 9, and MA.7.AR.2.1.

Working toward MA.912.AR.2.6 — one-variable linear inequalities including compound inequalities.

Where this sits on the test Last of three consecutive lessons on MA.8.AR.2.2, a **scored Grade 8 FAST benchmark** in Algebraic Reasoning. Math Nation's own teacher note calls the real-world problems “an extension of the benchmark” — that is not how the test treats them. Context items are the ordinary form of an Algebraic Reasoning question, and the reading is where the category loses its points.

This closes Unit 1. Ten lessons: seven on grade 7 codes that build toward the test, three on the first tested grade 8 benchmark. The vocabulary card and unit review are separate artifacts.

Materials Whiteboards and markers. Calculators — question 9 divides 12.50 by 3.49. Worksheet 7A.01.10. No cards to cut.

Lesson at a Glance

Windy Hill periods run **53 minutes**.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up: same context, different question	2	4 min
Notes I–IV: the phrase gives the symbol, building it, solving, the answer is a sentence	3–6	13 min
Example 1: a budget	7–9	5 min
Example 2: a minimum	10–12	5 min
Example 3: two answers, both 7.4	13–15	4 min
Whiteboards: 9 questions, each with its reveal	16–33	16 min
IXL — start today's skills	34	5 min
Total		53 min

Where the lesson lives: Notes IV (slide 6) and Example 3 (slides 13–15). Notes IV is the rounding rule with its reason attached; Example 3 is the same rule shown failing in both directions on real numbers. Everything else today is revision of Lessons 8 and 9.

Read this before you plan the day. Math Nation's components total 40 to 55 minutes — 5 for the Warm-Up, 10–15 for the Try It, 10–15 for Your Turn!, 10–15 for the Collaboration and 5 for the Wrap-Up — before any worksheet exists. At the top of that range it is over a full period with nothing written down. The plan above fits because the card-match Try It becomes the whiteboard sequence, and the Collaboration's two problems become Examples 1 and 2.

No partner or group work anywhere in this lesson. Math Nation has three, plus a suggestion to run the whole lesson as rotating centers: 1.10.2 is a partner card sort, 1.10.4 is a partner collaboration on both real-world problems, and the Collaboration tip asks students to verbalize with a partner before writing 2c and 2d. Both collaboration problems are now worked whole-class as Examples 1 and 2, and the card sort is two whiteboard rounds.

Vocabulary

at most — no more than — the value itself is allowed; $\leq$

at least — no less than — the value itself is allowed; $\geq$

maximum — the largest value a quantity is allowed to take

minimum — the smallest value a quantity is allowed to take

discount — an amount taken off a price

A note on the glossary. Math Nation lists no key terms and gives inequality and discount as additional terms. The four phrase entries are added because Notes I is built on them and because they, not the symbols, are what appears in FAST item stems.

Slide-by-Slide Teaching Notes

Slide 1. Inequalities in Real-World Contexts 1 min

- Post the learning target.
- Nothing new: this is Lesson 6's writing, Lessons 8 and 9's solving, and Lesson 7's interpreting, in one place.
- Say that — it is genuinely a consolidation day and students should hear it framed as one, with the assessment next.

Slide 2. Warm-Up 4 min

- TYPE: SSDD.
- INDEPENDENT.
- ANSWERS: 1) $4 + 2.50(6) = \$19$, arithmetic only. 2) $4 + 2.5m = 24$, so $m = 8$ miles exactly. 3) $4 + 2.5m \leq 25$, so $m \leq 8.4$ miles. Question 3 is today's whole lesson and most classes will write an equation for it — that is the moment to notice, not to correct.
- OFF THE SLIDE, YOURS TO SAY: One situation. Arithmetic, then an equation, then an inequality.

Slide 3. Notes: I. The phrase attached to the limit gives the symbol 4 min

- Students copy the table.
- The FAST uses all four phrasings and 'no more than' catches students out because it contains the word more.
- Read each row aloud as a sentence; the pairing of phrase to bar is the only thing this slide is for.
- OFF THE SLIDE, YOURS TO SAY: "At most" and "at least" include the number. "Fewer than" and "more than" do not.

Slide 4. Notes: II. Build it the same way as an equation, then add the symbol 3 min

- The reassurance is real and worth giving: everything they built in Lesson 6 still applies, and the only addition is one reading step.
- Students who found Lesson 6 hard will find this hard for the same reason, not a new one.
- OFF THE SLIDE, YOURS TO SAY: Same four steps as an equation, plus one: read the phrase that names the limit.

Slide 5. Notes: III. Solving does not change at all 3 min

- Short by design.
- There is nothing new here and saying so plainly is better than inventing content — the time belongs to Notes IV and to Example 3.
- OFF THE SLIDE, YOURS TO SAY: A context does not make an inequality behave differently. It only makes the answer mean something.

Slide 6. Notes: IV. The answer is a sentence, and it may need rounding 3 min

- Lesson 7's Notes IV, and the panel is the distinction that carries the points: the solution set and the answer are two different things and the assessment asks for both.
- A student who stops at 8.4 has solved and not answered.
- OFF THE SLIDE, YOURS TO SAY: " $m \leq 8.4$ " is the solution set. "8 whole miles" is the answer to the question.

Slide 7. Example 1 2 of the 5 min

- Whiteboards, inequality only. m = the number of miles traveled; $4 + 2.5m \leq 25$; the phrase is 'at most', which includes the 25.
- Hold out for the phrase — that is Notes I being used rather than recited.

Slide 8. Example 1 1 of the 5 min

- Say 'no flip' at both steps.
- The panel is a genuinely useful observation: costs and budgets almost never produce a negative divisor, so a flip in a money question is usually a mistake worth re-reading.
- OFF THE SLIDE, YOURS TO SAY: No flip. Nothing was divided by a negative — and money problems rarely do.

Slide 9. Example 1 2 of the 5 min

- Notes IV in full.

- The two arithmetic lines are the justification the assessment asks for; a bare '8' earns nothing, because there is no partial credit — the assessment's parallel item is keyed “a bare 46 with no arithmetic does not earn the point”.
- Ask why it is not 8.4 miles — because the meter charges by the mile and you cannot buy four tenths of one, which is a fact about taxis and not about rounding.

Slide 10. Example 2 *1 of the 5 min*

- Whiteboards, inequality only. b = the number of boxes opened; $50 + 24b \geq 170$.
- The phrase is 'at least', which gives $\geq$ and includes the 170.

Slide 11. Example 2 *2 of the 5 min*

- A deliberately clean answer.
- The panel is the habit: the question 'does this need rounding?' gets asked every time and sometimes answers itself, which is what stops rounding becoming automatic.
- OFF THE SLIDE, YOURS TO SAY: This one comes out whole, so no rounding decision arises. Ask anyway.

Slide 12. Example 2 *2 of the 5 min*

- THE ROUNDING CONTRAST.
- Same store, but the stock changes from 50 shirts to 60, so the inequality becomes $60 + 24b \geq 170$ and $110 \div 24 = 4.58$.
- It rounds UP to 5, because 170 is a minimum and 4 boxes falls short.
- Ordinary rounding would also say 5 here, which is why the next slide's example matters — it is the one where ordinary rounding gives the wrong answer.

Slide 13. Example 3 *1 of the 4 min*

- Take a vote.
- Ordinary rounding says 7 for both.
- Do not resolve it — the disagreement is the point, and this is the third time this unit that a decimal has rounded two ways, which is deliberate.

Slide 14. Example 3 *2 of the 4 min*

- The panel names why this is worth a slide: 7.4 rounds to 7 by the rule they have used since primary school, and 7 crates is the wrong answer.
- The symbol in the inequality tells you which way — $\leq$ down, $\geq$ up — and that is a rule they can actually use.
- OFF THE SLIDE, YOURS TO SAY: Ordinary rounding gets the second one wrong. The situation decides, not the decimal.

Slide 15. Example 3 *1 of the 4 min*

- The closing rule of the unit.
- It is short, it is memorable, and it converts Notes IV from a judgment into something checkable.
- Assessment question 12 is this exactly.
- OFF THE SLIDE, YOURS TO SAY: Last thing in the unit, and it is on the assessment. Read the symbol, then round.

Slide 16. Whiteboards • Question 1 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 17. Whiteboards • Question 1 of 9 — reveal *about 1 minute with the discussion*

- The baseline: a phrase to a symbol.
- Everything after this changes exactly one thing.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 18. Whiteboards • Question 2 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 19. Whiteboards • Question 2 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the phrase.
- The bar stays; the direction turns over.
- Notes I.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 20. Whiteboards • Question 3 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 21. Whiteboards • Question 3 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the bar is gone. 'More than \$25' does not include \$25 itself.
- Put questions 2 and 3 side by side — one word, and one number moves out of the answer.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 22. Whiteboards • Question 4 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 23. Whiteboards • Question 4 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: a whole situation instead of a phrase.
- Everything to the left of the symbol is Lesson 6, unchanged.
- Insist on the definition of m before the inequality.

- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 24. Whiteboards · Question 5 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 25. Whiteboards · Question 5 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: solve it.
- No flip — nothing was divided by a negative.
- A board that flipped saw a decimal and reached for the rule.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 26. Whiteboards · Question 6 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 27. Whiteboards · Question 6 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT CLOSES THE UNIT.
- One feature changed: answer the question rather than the algebra.
- Round DOWN, because $\leq$ is a maximum and 9 miles goes over.
- Ask for both arithmetic lines — the value that works and the next one that does not — because that justification is what the assessment points.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 28. Whiteboards · Question 7 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 29. Whiteboards · Question 7 of 9 — reveal *about 1 minute with the discussion*

- THE MIRROR OF QUESTION 6, and one feature changed: at most became at least, so the rounding goes the OTHER WAY. 5.71 rounds UP to 6 here and 8.4 rounded DOWN to 8 there, and the decimals are not what decided either one.
- THE BOARD TO LOOK FOR SAYS 5, from a student applying "5.71 is closer to 6" — or worse, getting the right answer for that reason, which is the board you cannot see.
- Ask for both arithmetic lines every time: 5 washes is \$175 and misses, 6 is \$210 and clears.
- Put questions 6 and 7 up together at the end of the round; they are the whole unit in two slides.

- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 30. Whiteboards · Question 8 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 31. Whiteboards · Question 8 of 9 — reveal *about 1 minute with the discussion*

- One feature changed from Lesson 9's round: there are now TWO flips, not one.
- Every question yesterday had exactly one, and a student who took that as the rule rather than as the count will stop after the first.
- THE BOARD TO LOOK FOR SAYS $a \geq -1$ — one flip, at the right place, and then a second division by a negative that went unnoticed at the end.
- Notes III's counting rule is the whole answer: count the divisions by a negative, and flip once each.
- Substitution settles it and takes ten seconds — $a = 0$ gives 2, which is not below -6 , so the $\geq$ version is out.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 32. Whiteboards · Question 9 of 9 — ask *about 90 seconds of writing*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 33. Whiteboards · Question 9 of 9 — reveal *about 2 minutes with the closing discussion*

- WRITTEN WORK, and the only question in the round that asks for it.
- With the worksheet gone this is the only place in the period the habit gets built, and IXL takes an answer without ever seeing the work.
- ANSWER: multiply by 4 first to get $10 \geq c - 3$, so $c \leq 13$.
- QUINN'S ERROR HAS A PRECISE NAME and it is not "she added 3": she added 3 to the left-hand side but only to the NUMERATOR on the right, and adding 3 inside a fraction over 4 adds only $\frac{3}{4}$.
- She did different things to the two sides while believing she had done the same thing.
- SAY WHAT SHE DID RIGHT TOO, because the question asks and because it is true — she never multiplied or divided by a negative, so leaving the symbol alone was correct, and her final line is a correct reading of her own third line.
- Check her answer to close it: $c = 22$ gives 4.75, and 2.5 is not greater than 4.75.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 34. IXL 5 min

- IXL.
- The skills are assigned in Google Classroom, not named here, and they are deliberately not ours -- Croix: "The students need questions that aren't exclusively mine."
- Makes them think." Circulate.
- The thing to watch for is a student answering in the box with no paper in front of them: IXL takes an answer and never sees the work, so this is the one part of the period where the habit can quietly lapse.

Worksheet 7A.01.10 — Answer Key

1. Quinn's error	<i>She added 3 to the left side but only to the numerator on the right.</i> Adding 3 inside the numerator adds $\frac{3}{4}$ to that side, not 3. The two sides were not changed by the same amount, so the statement stopped being equivalent.
2. note to Quinn	<i>Answers will vary — see the sample on the key.</i> A complete note names the specific error, gives a working alternative (clear the division first), and identifies something correct. Quinn never needed to reverse the symbol and she did not, which is worth telling her.
3. solve correctly	<i>$c \leq 13$ closed circle at 13, shade left</i> Multiply both sides by 4 first: $10 \geq c - 3$. Nothing reverses — 4 is positive.
4. $-b/4 + 5 > 1/2$	<i>$b < 18$ open circle at 18, shade left</i> $-b/4 > -4.5$, then multiply by -4 and REVERSE.
5. $8.8 \geq 0.8(-17.3 + 2.9x)$	<i>$x \leq 283/29$, about 9.76 closed circle between 9 and 10, shade left</i> Divide by 0.8 to get $11 \geq -17.3 + 2.9x$, then $28.3 \geq 2.9x$. The boundary has no tick mark; naming the two integers it falls between is part of the answer.
6. $1.5m - 10 < -1/4$	<i>$m < 6.5$ open circle halfway between 6 and 7, shade left</i> $-1/4$ is -0.25 , so $1.5m < 9.75$. Nothing reverses.
7. $-2/3(-3 - 12a) \leq -6$	<i>$a \leq -1$ closed circle at -1, shade left</i> TWO reversals: multiplying by $-3/2$, then dividing by -12 . Count them as you go — the answer is only backwards after an odd number of mistakes.
8. why $\leq$	<i>"At most \$25" allows exactly \$25 or any smaller amount.</i> > and $\geq$ point the wrong way; a strict < would forbid spending exactly \$25, which she is allowed to do.
9. solve Mary's	<i>$c \leq 3.58...$, so at most 3 cases of juice</i> $12.50 + 3.49c \leq 25$ gives $3.49c \leq 12.50$. Round DOWN — 3.58 is a ceiling. Three cases cost \$22.97.
10. four cases?	<i>No. $10(1.25) + 3.49(4) = \\$26.46$, which is more than \$25.</i> She is short by \$1.46. A bare "no" with no arithmetic earns nothing.

Question 5's boundary does not land on a tick, and that is deliberate. $8.8 \geq 0.8(-17.3 + 2.9x)$ gives $x \leq 283/29$, about 9.76. Rather than retuning the numbers, the item asks students to place the circle between 9 and 10 and name those two integers. Estimating a boundary onto a labeled

line is a FAST skill, and this is one of only three items on the lesson WORKSHEETS that requires ESTIMATING where the boundary falls between two ticks (the unit assessment and the practice test each carry two more, at questions 9 and 10) (Lesson 10's question 6 also lands off a tick, at exactly 6.5, but a midpoint is placed rather than estimated) — the others are Additional Practice question 4 here and Additional Practice question 9 in Lesson 8.

Question 7 reverses twice. Multiplying by $-3/2$, then dividing by -12 . Students who have absorbed “an even number of flips returns you to the start” will predict the direction before they finish, which is exactly the fluency Lesson 9 was building.

Question 10 is graded on the arithmetic. \$26.46 versus \$25. A bare “no” is not an answer to “justify your reasoning mathematically,” and the FAST scores it the same way.

Additional Practice 7A.01.10 — Answer Key

Post this in Google Classroom alongside the practice sheet. It is optional work for students who want more.

1. Jaxon's errors	(a) He divided by -4 without reversing. (b) He divided the right side by -4 twice — the factor out front AND each term inside. Both errors are needed; the question says there is more than one, and the item is one point.
2. note to Jaxon	Answers will vary — see the sample on the key. A complete note names both errors, gives a working alternative and identifies something correct. Choosing to divide rather than distribute was the right call.
3. solve correctly	$n > -10$ open circle at -10, shade right Divide by -4 and REVERSE: $-8 > -1.4n - 22$. Then $14 > -1.4n$, and divide by -1.4 and REVERSE again. Distributing instead gives $32 < 5.6n + 88$, which lands in the same place with no reversal at all — every step there divides by a positive number, and turning $-10 < n$ round into $n > -10$ is a rewrite, not a reversal.
4. $(5/3)(-a + 3/4) \leq -2$	$a \geq 1.95$ closed circle just below 2, shade right Multiply by $3/5$ to get $-a + 3/4 \leq -1.2$. The boundary sits a hair left of the 2 tick; saying “just below 2” is part of the answer.
5. $-3(2t - 4/3) > 6$	$t < -1/3$ open circle at $-1/3$, shade left Divide by -3 and REVERSE: $2t - 4/3 < -2$. The scale on this line runs in thirds so the boundary lands on a tick.
6. $(-6.2 + m)/4 > -3.3$	$m > -7$ open circle at -7, shade right The whole numerator is divided by 4, so multiply the whole other side by 4 first.
7. $(2/5)k - 8.4 \leq -4$	$k \leq 11$ closed circle at 11, shade left $2/5 = 0.4$ if a decimal is easier. Neither route reverses.
8. what $2/5$ means	Three fifths were sold, so two fifths of the original shirts are still on display. $(2/5)x$ is what remains of the original x; the 40 from the stock room is added to it.
9. solve	$x \geq 325$ $(2/5)x \geq 130$, then multiply by $5/2$.

10. what it means	<i>The store must have started with at least 325 shirts on display.</i> 325 is a minimum, so a decimal here would round UP — but 325 is already whole, so nothing changes. Not every reasonableness check ends in a change.
-------------------	--

Misconceptions to Watch

- **Distributing to the first term inside the parentheses only.** Named in Math Nation's list. Worksheet question 7 and Additional Practice question 1. (Math Nation TE)
- **Rounding a maximum up or a minimum down.** Notes IV, slides 13–15, worksheet question 9.
- **Reporting the decimal itself as the answer.** “3.58 cases of juice.” Worksheet question 9.
- **Choosing < when the context says “at most.”** Notes I and worksheet question 8.
- **Changing one side of an equation or inequality by a different amount than the other.** Quinn's error — adding 3 to a numerator instead of to the side. Worksheet questions 1 through 3.
- **Dividing a factored side by the factor and by the terms inside.** Jaxon's second error. Additional Practice questions 1 through 3.
- **Rounding an answer that was already whole.** Additional Practice question 10, where 325 needs nothing done to it.

Changes from the Math Nation Lesson

- Every answer number line is now labeled. Math Nation's Your Turn! prints eight graphs with unlabeled tick points, so a dot “at -5” is a dot anywhere and the item cannot be graded. Their own teacher tip suggests supplying numbered lines. Every graph on these sheets is scaled and labeled around the correct boundary.
- The rounding decision was made explicit and given a reason. Math Nation's key writes $m \geq 41.67$ with no comment, when the exact value is $41.\overline{6}$ and the rounding is upward — at \$41.66 a month Sasha reaches \$299.96 and misses. That contrast, paired against Mary's juice where 3.58 rounds down to 3, is now slides 13–15.
- Worksheet question 5 keeps its awkward boundary on purpose. $8.8 \geq 0.8(-17.3 + 2.9x)$ gives $x \leq 283/29$, about 9.76, which does not land on a tick. Rather than retuning the numbers, the item asks for placement between 9 and 10 and for those integers to be named.
- Worksheet question 6 was rewritten as $-1/4$ rather than a fraction printed with -4 in the denominator.
- Jaxon's error analysis now says there is more than one error. As printed it asks “what error did Jaxon make?” when his work contains two: he divided by -4 without reversing, and he divided the right side by -4 twice. Students who find only one had answered the question as asked.
- Every partner and group activity is replaced — three of them, plus a suggestion to run the whole lesson as rotating centers. The card sort is the whiteboard sequence; both Collaboration problems are Examples 1 and 2, worked whole-class.

- The real-world problems are not treated as an extension. Math Nation's teacher note calls them "an extension of the benchmark." On the Grade 8 FAST, context items are the ordinary form of an Algebraic Reasoning question, and this teacher's edition says so.
- The Wrap-Up was replaced. Math Nation's is "write an explanation to a classmate about how solving an inequality is different from solving an equation," keyed "answers will vary" — the fourth wrap-up in the unit that produces no gradeable mathematics. The Ticket Out the Door has students write and solve an inequality, justify a boundary that is deliberately exact, and translate one phrase into a symbol.
- Number lines are printed for every graphing item on both sheets.
- Answer keys were built for both sheets from the same source file as the student versions, so the two cannot drift apart.
- Nothing in the mathematics was changed.